Shortcut to adiabatic population transfer in quantum three-level systems: Effective two-level problems and feasible counterdiabatic driving

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Abstract

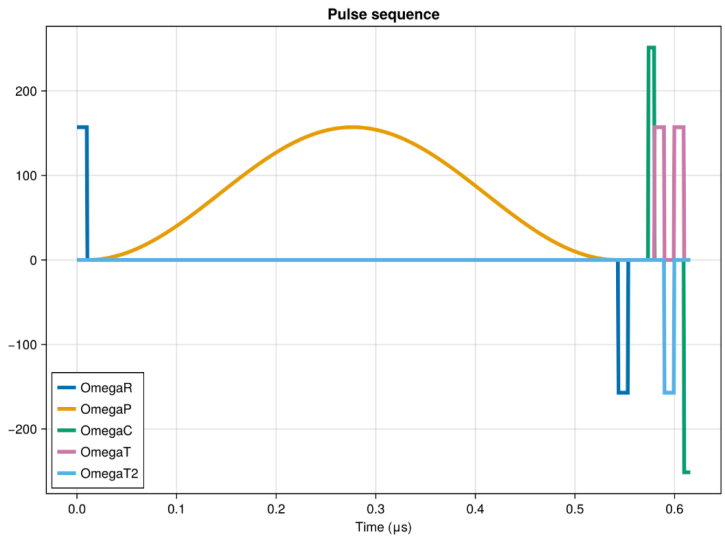
In this note, I intend to change the pulse shape of Ω_p in Y-type energy level systems and see that if the laser pulse can be optimized

Keywords. measurement-free error correction · multi-species array · parity-gate · multi-qubit gate · etc

AMS subject classification. 65M70, 65M12

0.1 Introduction

In this section, the original pulse sequence for Y-type system is shown below:



It is obvious that the pulse duration of Ω_p is much longer than other pulses, which will induce decay error and decoherence. Thus, we intend to speed up the pulse Ω_p.

Initially, we try to use DRAG method to eliminate the leakage to intermediate state. However, during the investigation process, I found that the DRAG method in Rydberg atom system is mainly targeted at decreasing the leakage to nearby Rydberg states and most relevant papers are focusing on two-level systems.

Thus, I try to use the counterdiabatic driving method to speed up the adiabatic process.

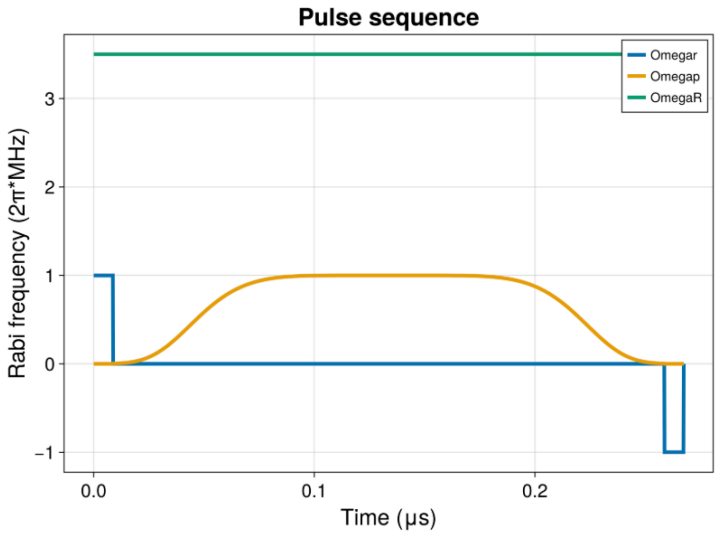
There are mainly two different conditions: the first is transition-blocking process: I concentrate on the EIT process considering ground state $|0>, |1>$, intermediate state $|p>$ and Rydberg state $|r>$. In this process, there are mainly two different ways to transition-blocking process: 1)Use adiabatic driving to maintain the system in the dark state; 2)Use counterdiabatic driving to cancel the non-adiabatic transition.(This is still under investigation)

The second is population-transfer process with Raman laser. In this case, we cannot use counterdiabatic driving to compensate the non-adiabatic transition directly since the two coupling lasers(Ω_p) are of same pulse shape. So, we use the effective two-level model to couple two ground states $|0>, |1>$. The effective coupling strength is $\Omega_{\text{eff}} = \frac{\Omega_p^2}{2*\Delta}$ and the effective detuning is $\Delta_{\text{eff}} = 0$. The pulse area of Ω_{eff} should be π to realize the population transfer.

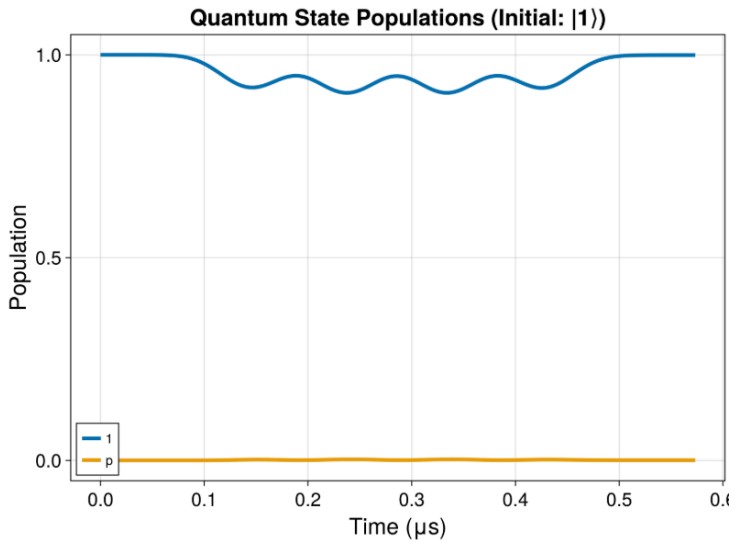
We first changed the pulse shape of Ω_p to Gaussian shape to see the results:

0.2 Simulation

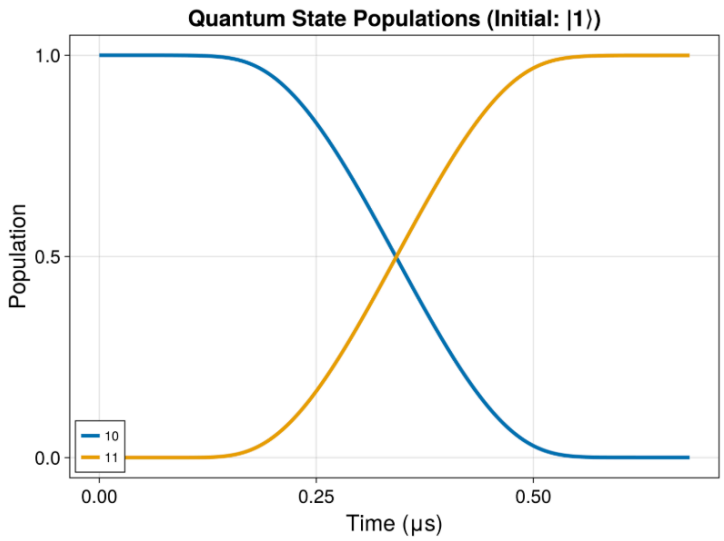
The new pulse shape is shown below:



When control qubit is in state $|0>$, target qubit is invariant:



When control qubit is in state $|1>$, target qubit is transferred from $|0>$ to $|1>$:


Quantum State Populations (Initial: |1⟩)
1.0
0.5
0.0
Population
0.00
0.25
0.50
Time (μs)
|0⟩
|1⟩